Assignment 6

https://www.notion.so/Assignment-6-219ee40c6a0880d5af29c7f66c5704d9?source=copy_link

1. What is the output of the following code?

```
x is 60
The size of numbers is 30
```

2. What is the array index type? What is the lowest index? What is the representation of the third element in an array named a?

```
integer
0
a[2]
```

3. Which of the following statements are valid?
    a. invalid
    b. valid
    c. invalid
    d. invalid
    e. invalid
    f. invalid
4. Write statements to do the following:
    a. double d[] = new double[10];
    b. d[d.length - 1] = 5.5;
    c. System.out.println(d[0] + d[1]);
    d.
    ```
    double sum = 0;
    for (int i = 0; i < d.length; i++) {
    ```

```
    sum += d[i]
};
```

e.
```
double min = d[0];
for (int i = 1; i <d.length; i++) {
    if (d[i] < min) {
        min = d[i];
    }
}
```

f.
```
int random = (int)(Math.random() * d.length);
System.out.println(d[random]);
```

g.
```
double[] a = {3.5, 5.5, 4.52, 5.6};
```

5. What happens when your program attempts to access an array element with an invalid index?

```
ArrayIndexOutOfBoundsException
```

6. Figure 1 shows the distribution of flood cases that happened in five states in Malaysia. Based on Figure 1, answer the questions below:

a.
```
int[][] floodCases = {
{2,13,5},
{3,25,9},
{6,11,12},
{16,20,45},
{17,22,5},
}
```

b.
```
int total = 0;
for (int i = 0; i < floodCases.length; i++) {
   for (int j = 0; j < floodCases[i].length; j++) {
      total += floodCases[i][j];
 }
}
System.out.println("Total number of flood cases: " + total);
```

c.
```
for (int i = 0; i < floodCases.length; i++) {
   int max = floodCases[i][0];
      for (int j = 1; j < floodCases[i].length; j++) {
         if (floodCases[i][j] > max) {
            max = floodCases[i][j];
    }
   }
   System.out.println("The highest flood case in State " + i + " is " + max)
}
```

7. What is the output of the following code?

```
111111
```

8. Suppose the following code is written to reverse the contents in an array, explain why it is wrong. How do you fix it?

```
i < list.length is wrong, change to i <j
```

9. Show the output of the following two programs:

a.
```
number is 0 and numbers[0] is 3
```

b.
```
2
```

```
4
```

10. Universiti Putra Malaysia (UPM) has a golf course that is open to public, however, membership is only for the staff. Assume that there are seven members of UPM Golf Club with the membership number is stored in an array named memberID as follows:

```
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        int[] memberId = {10011, 10012, 20011, 20012, 10013, 10014, 20013};
        int[] daysPlayed = readInput(memberId.length);
        double[] fee = calculateFee(daysPlayed);
        displayOutput(memberId, daysPlayed, fee);
    }

    public static int[] readInput(int numberOfMembers) {
        Scanner scanner = new Scanner(System.in);
        int[] daysPlayed = new int[numberOfMembers];

        System.out.println("Enter number of days played for each member:");
        for (int i = 0; i < numberOfMembers; i++) {
            System.out.print("Member " + (i + 1) + " - Days played: ");
            daysPlayed[i] = scanner.nextInt();
        }
        return daysPlayed;
    }

    public static double[] calculateFee(int[] daysPlayed) {
        double[] fee = new double[daysPlayed.length];
        for (int i = 0; i < daysPlayed.length; i++) {
            double gameFee;
            if (daysPlayed[i] <= 5) {
                gameFee = 100.00;
```

```
            } else if (daysPlayed[i] <= 10) {
                gameFee = 200.00;
            } else if (daysPlayed[i] <= 15) {
                gameFee = 300.00;
            } else {
                gameFee = 500.00;
            }
            fee[i] = 150.00 + gameFee;
        }
        return fee;
    }

    public static void displayOutput(int[] memberID, int[] daysPlayed, double[]
        System.out.println("Golf Club Monthly Report");
        System.out.printf("%-10s %-15s %-15s\n", "MemberID", "Days Played",
        System.out.println("-----------------------------------------");
        for (int i = 0; i < memberID.length; i++) {
            System.out.printf("%-10d %-15d RM%-14.2f\n", memberID[i], daysPla
        }
    }
}
```